

Hand Gesture Authentication Using Optimal Feature Selection and Dynamic Time Warping based K-nearest Neighbor

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ABSTRACT

Network security is becoming a critical issue due to the spread of computer networks to ordinary people. User authentication is one of the most important part for network security. Although passwords are used in many personal computers, it has many well-known problems and it is antiquated. On the other hand, fingerprint or face based authentication has the risk of information leakage by reproducing fingerprint or face from the photograph. In this paper, we have designed a hand gesture-based user authentication using leap motion to overcome of information leakage problem. We have collected data from 25 participants and used the data to analyse the accuracy of our authentication system. The experimental results are also measured by FAR (False Acceptance Rate) and FRR (False Rejection Rate). At first, Feature selection has been used to select important feature out of 246 features. Finally, Dynamic Time Warping (DTW) based K-Nearest Neighbor (KNN) Classifier has been used to authenticate a person. Our system has produced 96.73% accuracy in user authentication. It is also showed 1.25% FAR and 3.62% FRR. Our results indicate that proposed hand gesture with Leap Motion can be viable authentication approach.

CCS CONCEPTS

• Networks; • Network properties; • Network security;

KEYWORDS

Authentication, Leap Motion, Hand Gesture, Feature Selection, Dynamic Time Warping, K-Nearest Neighbor

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1 INTRODUCTION

User authentication is a critically important task in cybersecurity. Nowadays, password and biometrics are generally used for authentication. In particular, notwithstanding text-based passwords are a traditional method, we suffer from many well-known problems [1]. The strong passwords are difficult to remember because of the complex structure. Moreover, we have not to reused passwords. We don't prevent leak passwords from the company by cracking. On the other hand, fingerprints identification or face recognition are widely used biometrics on mobile phones. Although fingerprints and face are unique, the attacker can create bogus fingerprints or face by the photograph of your finger or face. As a result, the attacker can do those recognition methods instead of you. Also, it is important to notice that fingerprint recognition method necessary touching to the device.

This research set out to resolve those problems using hand gesture identification. This way has a low risk of leaking information because we reveal a person's characteristic from the hand gesture. In addition, this method does not require touching to the device owing to do hand gesture in mid-air. This type of authentication is based upon moving patterns of hand detected in an individual, making authentication not only easier but something that can be done without even having to think about it. We implemented the method by using a 3D motion sensor called the Leap Motion controller.

The Leap Motion sensor captures user's hand actions above the camera and translate them into 3D input using the two infrared cameras and the infrared LED within the sensor [2]. Recently, there has been research on camera-based methods using the Leap Motion controllers. Satoru et al. [3] has used seven types of 3D gestures to do user authentication. For all the gestures, they have got the true acceptance rates more than 90 %, and the equal error rates less than 4%. Chan et al. [4] used a circle-drawing gesture captured by leap motion with one finger for biometric authentication. Saritha et al. [5] also developed the Leap Motion controller based authentication system to treat circle-drawing, swipe, screen-tap, and key-tap gestures. 3D hand gesture signatures based system has been developed by Xiao et al. [6] using the Leap Motion controller. Wong and Kang [7] implemented a biometric authentication method by using the Leap Motion controller. They used stationary hand gestures that consists of free motions of fingers without the motions of hands, wrists, and arms. Chan et al. [4], Saritha et al. [5], and Xiao et al. [6] used relatively simple hand gesture because they did not apply gestures with the complex motions of fingers. Although Wong and Kang [7] used more complex stationary hand gesture, they still did not use the motions of hands.

Normally, the Leap Motion controllers based system share the same motivation as our method. However, our method is more

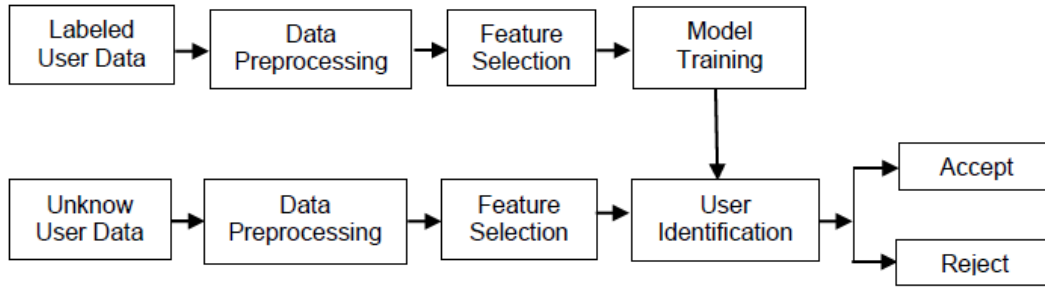


Figure 1: Process of User Authentication in our study.

focused on the use of an easy to remember but reliable 3D hand gestures. We expect individual differences would appear greatly by measuring the hand gesture that moves all the fingers. In other words, the gesture should be dynamic. Moreover, we notice the problem that tracking is incorrect when fingers hide from Leap Motion camera. As a result, a subject should not be hidden from the Leap Motion camera because the Leap Motion provides ambiguous data. In this research, a new hand gesture has been devised that moves all fingers without hiding any fingers. Primarily, we have collected 246 features from the leap motion controller and then we have selected 27 features depending on their importance. Finally, a DTW based KNN model has been trained to do user authentication using the selected features. From this study, we have confirmed the possibility that the proposed hand gesture is useful to user authentications.

2 PROPOSED MODEL

The process of our system is shown in Figure 1. There are two important phases in our system. One is training phase and another is authentication phase. The main purpose of the training phase is to train the model using labeled data. After that, during the authentication phase the trained model will be used to identify whether the unknown data is from the legitimate user or from an impostor.

3 DEFINED HAND GESTURE AND DATA COLLECTION

3.1 Measurement Device

We have measured the hand gesture using Leap Motion. Leap Motion is a device that captures movements of hands by infrared cameras. In recently, Leap Motion is in demand as a controller in VR. Figure 2 shows the recognition of Leap Motion; the hand is displayed as a skeleton. Basically, we can get the data of the points of the hand using API of the Leap Motion Controller (Leap Motion software). Leap Motion is a high-performance device, according to the official, Leap Motion tracking accuracy is 0.01 mm. whereas, according to previous research, the accuracy is 0.2 mm for normal accuracy and 1.2 mm for the dynamic gesture [8]. Leap Motion photographs hands and fingers lit up by the infrared LEDs by two monochromatic IR cameras and sends about 120 frames per second to the host computer. It internally has a standard model of human

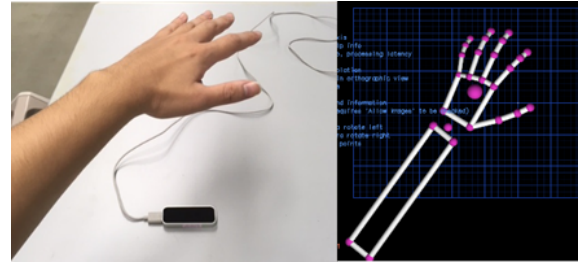


Figure 2: Hand gesture recognition using Leap Motion

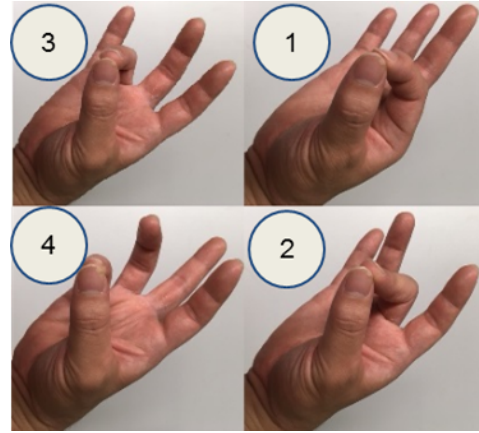


Figure 3: Flow of gesture

hands, and treats various motions of hands by associating sensor data with the internal model. However, it cannot accurately sense fingers that are hidden by other fingers or by palms.

3.2 Used 3D Hand Gestures

To do biometric authentication using the geometries and the 3D gestures of a hand, we need to define a reliable gesture pattern. In this work, we have devised a hand movement gesture obtain the geometry and the position of the hand in real time using leap motion. We have defined a specific order in the flow of movement of finger in our proposed hand gesture which is shown in Figure 3.

Table 1: List the Name of Different parts of a hand, corresponding feature type and number of features

Part of the Hand	Feature Type	Number of Feature
Fingertip, Bone Pre-Joint, Palm, Wrist	Co-ordinate Position	81
Finger, Bone, Hand	Length, Width	51
Fingertip, Hand	Visibility	6
Fingertip, Bone, Palm	Velocity, Direction (The Direction of the palm is the Roll, Pitch, and Yaw angles centered on the palm)	96
Hand	Grab angle, Grab strength, Pinch distance, Pinch strength	4
Palm	Normal (The normal vector to the palm),	3
Fingertip	Extended (A real number between 0~1 indicates whether the finger is open or not.)	5

3.3 Data Collection Procedure

We measured hand gesture against 25 subjects (10 to 20 years old). The measurement is the right hand irrespective of dominant hand. We made some rules for measuring gestures. The description of the rule are as follows: (i) The position of hand as directly above Leap Motion; (ii) Keep 15cm or more and less than 40cm from Leap Motion; (iii) The measurement starts after opening the hand; (iv) Open hands after measurement ends; (v) Subjects perform hand gesture naturally; (vi) Measure hand gesture 20 times. Leap Motion’s camera is facing directly above. As a result, we should consider about gestures that all fingers do not hide from cameras. As a result, we determine the hand gesture that taps the fingers with each finger. We expect that it is easy to detect the feature because all the fingers are moving.

3.4 Dataset Formation

In our data collection process, 25 participants are involved. Each participants or person gave hand gesture 20 times. As a result, the final dataset contained 500 gesture samples.

4 FEATURE DESCRIPTION AND PREPROCESSING

4.1 Feature Description

We have collected 246 features from the hand geometry which has been captured by Leap Motion. The Table 1 showed the list of hand parts and their corresponding features.

4.2 Normalizing Hand Gesture Data

In our work, we have used the x, y, z coordinate position of finger tips, bone’s prejoint, wrist and palm location. Normally, every time a user cannot do his hand movement by keeping the hand always in the same location. We may get different data for x, y, z coordinate position of finger tips, bone’s prejoint, wrist and palm location for the same user. To overcome this problem, we have normalized our data (only x, y, z coordinate position) based on the position of the palm by performing transformations; i.e., moving the palm to the origin.

5 AUTHENTICATION USING DTW BASED KNN

5.1 K-Nearest Neighbor (KNN) Classifier

The K-Nearest Neighbor (KNN) algorithm is applied to authenticate the unknown person in our system. The KNN algorithm make a decision based on the idea that if majority of the K nearest neighbors of an unknown data are belong to a certain category, then the unknown data will be classified into that category [9].

KNN algorithm uses Euclidean distance most frequently as distance measurement method. However, two sequences must have equal length of values to calculate the Euclidian distance. At the same time, the values of the sequences must be one-to-one. Obviously, this distance measurement method does not apply to our research, because it is not possible to calculate distance between two the sensor data with variable length. We have used the DTW (Dynamic Time Warping) algorithm as the distance method to overcome this limitation.

The calculated DTW distance will be used in the classification or authentication process of KNN algorithm. Normally, we consider the value of K is equal to one. In our system, we have tuned the value of K. The steps of DTW-1NN algorithm for authentication is given below:

- Compute the DTW distance between all labeled samples and the unknown sample.
- All computed DTW distances are sorted and the labeled sample with the smallest distance is taken as the nearest neighbor of the unknown sample.
- Set class label of the nearest neighbor is the class label of the unknown sample.

5.2 Calculation of Dynamic Time Warping (DTW) Based Distance

Dynamic Time Warping is used to compare the similarity or calculate the distance between two sequences of different length. Assume we have two-time sequences, X and Y, of length n and m, respectively, where $X = \{x_1, x_2, \dots, x_n\}$ and $Y = \{y_1, y_2, \dots, y_m\}$ and we want to compute the matching cost or distance: DTW (X, Y). The distance is computed based on dynamic programming using the

following formulation [10]:

$$D(i, j) = d(x_i, x_j) + \min \begin{cases} D(i, j-1) \\ D(i-1, j) \\ D(i-1, j-1) \end{cases} \quad (1)$$

where the distance function $d(\cdot, \cdot)$ varies with the application. In our gesture recognition system, $d(x_i, y_i)$ is defined as,

$$d(x_j, y_j) = (x_i - y_i)^2 \quad (2)$$

and consequently,

$$DTW(X, Y) = D(n, m) \quad (3)$$

6 FEATURE SELECTION

Feature selection is the method to reduce the number of irrelevant input features for developing a predictive model. Normally, it reduces the number of input variables to decrease the computational time of the model [11] and to increase the performance of the model in some cases [12]. In our solution, we have produced individual feature importance depending on accuracy using the following algorithm:

- i. Select one feature from 246 features.
- ii. Find identification accuracy using DTW based KNN for this feature.
- iii. Repeat steps i and ii for all (246) features
- iv. Sort those feature according to accuracy
- v. Select those features which produce more than 80% accuracy.

7 PERFORMANCE EVALUATION METRICS

In the testing procedure, we have chosen accuracy to assess the classification performance of KNN as our performance metrics. Moreover, we have also calculated another two metrics: FAR (False Acceptance Rate) and FRR (False Rejection Rate) to measure authentication performance of our system. FAR is the rate of accepting an imposter biometric sample as a legitimate user. On the other hand, FRR is the rate of rejecting a valid user incorrectly as if the user is an imposter. In an authentication system, larger FAR value

is more harmful compared to FRR values, because FAR allows imposter to easily access the system. For the authentication system, we expect that it can effectively stop the attack of impostors, and at the same time, it can give easy access to the legitimate user. So, the smaller values of FAR and FRR represent the better authentication performance of the system.

Our final classifier is a multi-class classifier and it gives a label that represents the predicted name of a user. The general strategy in evaluating the FAR and FRR errors for multi-class classifiers is to reduce the problem of multiclass classification to multiple binary classifications in which each class has its own value of FAR and FRR errors. In our case, we have calculated the average FAR and FRR from all classes.

8 EXPERIMENTAL RESULTS

8.1 Experimental Setup

All 25 subjects produced a total of 500 gesture samples. We have used 70 percent samples to train the model and 30 percent samples to test the model. We have also used these 70 percent training samples to find the important features.

8.2 Important Selected Features

We have selected 27 features out of 246 features using feature selection technique defined in section 6. The selected feature is listed in Table 2

8.3 Results and Discussions

In our system, KNN has been used to classify the legitimate user and the imposter. Here, we have used $K=3$. Table 3 summarizes the performance of our system. To show the performance of the proposed method, we have calculated accuracy, the average FAR and FRR. Our system produced 96.73% accuracy in user authentication. The FAR and FRR are 1.25% and 3.62% respectively of our system.

Table 4 illustrates the results of our research and some similar previous research works. All researchers used different types of hand gesture to authenticate a person. They also collected data

Table 2: List of Selected Features

SN	Feature name	SN	Feature Name
1	Thumb_Finger_Intermediate_Direction_x	15	Ring_Finger_Tip_Velocity_x
2	Index_Finger_Tip_Direction_x	16	Ring_Finger_Tip_Velocity_y
3	Index_Finger_Intermediate_Direction_x	17	Ring_Finger_Intermediate_Direction_y
4	Index_Finger_Metacarpal_Direction_x	18	Ring_Finger_Metacarpal_PreJoint_x
5	Index_Finger_Proximal_Direction_y	19	Ring_Finger_Metacarpal_PreJoint_z
6	Middle_Finger_Tip_Direction_x	20	Ring_Finger_Metacarpal_Width
7	Middle_Finger_Tip_Direction_z	21	Ring_Finger_Proximal_Prejoint_z
8	Middle_Finger_Tip_Velocity_x	22	Little_Finger_Tip_Direction_z
9	Middle_Finger_Tip_Velocity_y	23	Little_Finger_Tip_Velocity_y
10	Middle_Finger_Metacarpal_Direction_x	24	Little_Finger_Proximal_Direction_y
11	Middle_Finger_Proximal_PreJoint_z	25	Wrist_Position_y
12	Middle_Finger_Proximal_Direction_x	26	Wrist_Position_z
13	Middle_Finger_Proximal_Direction_y	27	Hand Width
14	Ring_Finger_Tip_Position_x		

Table 3: Accuracy, FAR and FRR of our system

Accuracy	FAR	FRR
96.73%	1.25%	3.62%

Table 4: Compare results of previous research

Authors	Methods	Performance
Saritha et al. [5]	DTW	Circle: 60%; Swipe: 70%; Screen Tap: 90%; Key Tap: 85%
Ilhan Aslan et al. [13]	DTW	Accuracy: 86-90%
Aman Chahar et al. [14]	NB, NN, RDF and Fusion	NB: 75.78%; NN: 78.55%; RDF: 78.04%; Fusion: 81.17%
Jinghao Zhao et al. [15]	HMM	Accuracy: 95%; FAR: 1.65% FRR: 4.82%
Our Proposed	DTW and KNN	Accuracy: 96.73%; FAR: 1.25%; FRR: 3.62%

using Leap Motion. From Table 4, it is clear that, our research obtained higher accuracy with lower FAR and FRR. The performance improvement of our system comes from two aspects. First, we have devised a hand gesture that can capture the user characteristics very accurately. Second, we have used important feature through feature selection mechanism.

9 CONCLUSION

Cash cards, password or biometrics information based user authentication systems is vulnerable to external attacks. To improve the security, a touch-less authentication system has been proposed in this work. This paper has presented the design and evaluation hand gesture-based authentication system using the leap motion sensors. As shown in the experimental results, the system designed in our study can produce accuracy 96.73% with average FAR 1.25% and FRR 3.62%. Our findings indicate that proposed hand gesture-based user authentication can provide a significant enhancement for traditional authentication systems.

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